NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.6c Wind-field reconstruction fusing observations and simulation

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:36 — 2026-10-15 02:32

Stage conclusion

Project report

H Observation–Simulation Fusion for Wind-Field Reconstruction

Objective

Correct and reconstruct simulated surface wind fields using experimental and observational data (sparse station observations → continuous wind field). Step 1: literature review + leave-one-station-out benchmark on synoptic non-storm winds (Ordinary Kriging / background field + residual Kriging / machine learning) to quantify how much multi-source fusion beats strong interpolation baselines and whether it is worth continuing. Subsequent phases extend to typhoons/hurricanes, extratropical cyclones, and thunderstorms, providing reconstructed wind fields as an independent wind-speed input to Node I (wind-damage loss vs wind-speed database).

Current Status

Wrapped up on 10-14 (Shelved). Final deliverable: *Station Wind-Speed Reconstruction Protocol v1.2* (handed over to Node I): on the ASOS station network, `background + residual Kriging` improves over pure Ordinary Kriging by only `3–6%` (`~0%` for thunderstorms), reaching conditional utility only at sparse station spacing (~80 km) or high-wind hurricane stations; for hurricane event peaks, `Holland + residual` eliminates peak low bias, while hazard-stratified additive bias correction removes peak low bias across the remaining hazard classes (coarse estimate for thunderstorms). For thunderstorms, station interpolation underpredicts peaks by `3–5 m/s`, and both low-elevation single-Doppler radar trials (`H6` and `H7`) proved "inconclusive" and were wrapped up under the stop-loss rule as "infeasible under current public data constraints". **Why Shelved**: Station reconstruction and peak correction across all four hazard classes are complete and delivered to Node I, while two targeted radar pilots on the thunderstorm track yielded no positive signal. **Restart Conditions**: Node I uncovers a systematic flaw when applying the protocol; near-field surface observations become available to validate low-level radar winds inside thunderstorms; TDWR Level-II Nyquist unfolding is reliably resolved with stations sitting under `<300 m` beams; or official verification clarifies NWS DAT wind-speed units and estimation protocols.

Compute and Storage Boundaries (Updated 10-15 per Liang Hu's 10-14 Tightening)

- **No compute or compute-data storage on the workstation**: If restarted, all solvers, batch leave-one-out runs, radar decoding, and large downloads must run and reside on CRC/TACC or Colab; the workstation holds only scripts, compact results (<200 MB per item), figures, and reports, with temporary staging >1 GB requiring prior parent-node approval. New large datasets prioritize TACC (CRC group storage is ~95% full), and CRC jobs default to single-node.
- Legacy compute data from this node have been archived to TACC by the root Lead and will be read from TACC if needed for future audits without pulling them back to the workstation. Earlier "0 CU local workstation run" clauses in the `H1–H7` briefs are superseded.
- No per-trial CU cap on Colab; pause and report to the parent node if a single trial is projected to exceed 100 CU; always run `colab stop` upon completion and dispatch only one execution agent per round.
- Cluster jobs require a ComputeOps resource request and registry entry, capped at ≤50 SU per job with separate pre-check and submission commands.

Hierarchy Rules (Set by Root Lead on 10-14, 5-Layer Cap Post-Migration)

- Do not open Level-3 sub-projects under this Level-2 node; handle subdivisions via tasks inside this node or escalate to the parent node to open a sibling Level-2 node.

- Only two delegation hops remain below this node: `Lead → bot → tool`; briefs must not permit tool bots to re-delegate.
- Subtree listings follow the UI project tree rather than grouping by parent node.

Decisions Made

- 2026-10-14 Step 1 design and evaluation criteria →
- 2026-10-14 **Wrap-up**: Protocol `v1.2` finalized as final deliverable; near-surface thunderstorm wind reconstruction declared infeasible with current public data →
- 2026-10-14 `H7` final pilot and stop-loss rule →
- 2026-10-14 `H6` thunderstorm radar pilot: inconclusive →
- 2026-10-14 `H5` verdict and protocol upgrade to `v1.2` →
- 2026-10-14 Protocol `v1` review verdict and release of `v1.1` →
- 2026-10-14 Finalize thunderstorm station test, wrap up leave-one-out phase, and submit Protocol `v1` for review →
- 2026-10-14 Finalize extratropical cyclone phase and launch `H4` thunderstorm pilot →
- 2026-10-14 Finalize hurricane phase and advance to extratropical cyclones →
- 2026-10-14 Wrap up synoptic non-storm phase and advance to landfalling hurricanes →
- 2026-10-14 `r1` verdict ("interpolation suffices") and pivot in `r2` to thinned station networks →

Established Premises (In-Line Data)

- Pure interpolation baselines are strong: in the Lab's TPU wind-tunnel trial, ML beat interpolation on `0/5` strict interpolation splits; in Han 2026, pure Kriging achieved MAE `0.36–0.37 m/s`, outperforming LBM-LES at `0.46 m/s` (digital-twin wind-tunnel card, p. 39 Table 6).
- Generative data assimilation (`GenDA`) has been demonstrated only on 2D steady RANS slices of a single idealized urban block with synthetic sensor nodes and closed data.

Trials and Results

Date	What Was Done	One-Line Result
2026-10-14	Grok quick search + primary source verification by Lead	Verified real-station precedents: `KiloDA` (Shanxi complex terrain, 238 stations held out as a spatial block from 1,340 stations) reduced RMSE by `15.4%` over Kriging with `89.5%` of stations improved; Kiessling 2021 (Sweden) beat Universal Kriging by ~`5%`; synthetic twin-experiment gains (`49%`) excluded from real-station evidence; `S2G` numbers pending check

Date	What Was Done	One-Line Result
2026-10-14	r1 Synoptic non-storm leave-one-station-out (research assistant; 137 stations, 31,898 station-hours)	OK 1.613, IDW 1.634; ERA5 + residual OK 1.553 (−3.8%, 81/137); GBT + residual 1.524 (−5.5%, 87/137); in the >50 km bin, OK 2.142 vs GBT + residual 1.625; peaks broadly underpredicted by ~1.5 m/s → interpolation suffices
2026-10-14	H7 Final thunderstorm radar pilot (2 Chicago cases, KLOT low beam + 108 DAT points; research assistant)	Test A had 0 qualified station events (beam height 385–736 m over stations); in Test B (assuming DAT in mph), median |radar − DAT| was 16.0 m/s vs 9.7 m/s for |nearest station − DAT| → inconclusive, triggering stop-loss
2026-10-14	H6 Thunderstorm radar near-surface feasibility (2 Dallas days, KFWS lowest tilt; research assistant)	Across 18 station-events: sustained peak bias shifted from −3.59 (interpolation) to −3.28 m/s (radar) while RMSE worsened from 4.71 to 5.18 m/s; beam height 258–1,646 m AGL; 0 DAT thunderstorm points on those two days; TDWR unused → inconclusive
2026-10-14	H5 Peak underprediction correction (leave-one-event-out across all 4 hazard classes; research assistant)	Hazard-stratified additive correction reduced peak RMSE by −12% (synoptic), −22% (extratropical), −25% (thunderstorm sustained), and −30% (thunderstorm gust), bringing pooled bias within ±0.5 m/s → effective (per-event thunderstorm residual remains ±2–3 m/s; hurricane Holland + residual needs no correction; single cross-hazard offset fails)
2026-10-14	H4 Thunderstorms (Dallas 4 gradable days, 44 1-minute stations; research assistant)	Peak sustained wind IDW 4.78 → ERA5 + residual 4.79 m/s (−0.2%) → interpolation suffices; peak bias is −3.2 m/s (sustained) and −5.1 m/s (gust), vs −8.8 m/s for standalone ERA5 (very small sample)
2026-10-14	H2 Supplement: adding Hurricane Francine (aligned with B)	On Francine alone, primary metric is −22.8% (25/44, conditional) and strong-wind 10 stations −52.8%; pooled across 10 events + Francine, overall is −3.8% (interpolation suffices) and independent strong-wind stations −10.8% (conditional) → hurricane verdict unchanged

Date	What Was Done	One-Line Result
2026-10-14	H3 Extratropical cyclones (South-Central Plains, 4 strictly compliant events / 359 stations; research assistant)	Peak RMSE OK 2.30 → ERA5 + residual 2.22 m/s (−3.2%, 207/359) → interpolation suffices (5-event sensitivity −3.2%, 2-event −5.5%; ks_20180306 had 1 VCTS report; standalone ERA5 underpredicts peaks by 3.2 m/s, interpolated methods by 1.2–1.4 m/s)
2026-10-14	H2b Strong-wind secondary criterion (9 independent events, 273 stations with ≥12 m/s)	Kriging 3.89 → Holland + residual 3.50 m/s (−10.0%, 154/273) → conditional (only Zeta passed both thresholds individually) → hurricane phase finalized
2026-10-14	H2 Cleaned land-station cohort (10 events, 745 stations; research assistant)	Peak RMSE OK 2.79 → ERA5 + residual 2.70 m/s (−3.2%, 379/745) → interpolation suffices (confirming the uncleaned −15% came from offshore/coastal-water stations); Holland + residual reduces peak bias from −1.18 to ~0 (−11% at 1–3 RMW, worse inside <1 RMW)
2026-10-14	r2-clean (cleaned station list, hourly variogram refit on residuals; research assistant)	Full-network r1 −2.6% (78/134); thinned to 40 km −6.2%, 54 km −9.4% (interpolation suffices), and 82 km −12.5% (124/134, conditional); GBT still trails ERA5 + residual → synoptic phase finalized
2026-10-14	H2 10-event raw cohort (same cohort as B v1; research assistant)	Uncleaned pooled peak RMSE OK 3.21 → ERA5 + residual 2.73 m/s (−15.0%, 413/794 stations, superseded by cleaned land cohort above); standalone frozen Holland trailed ERA5 on most storms (overpredicting Ida by +7.3 m/s)
2026-10-14	H2 Initial pilot: Beryl 2024 + Nicholas 2021 leave-one-out (research assistant)	Uncleaned 201 stations: OK 3.17 → Holland + residual 2.46 (−22.4%) and ERA5 + residual 2.53 m/s (−20.3%), improving at 105/201 stations
2026-10-14	r2 Thinned station networks (50/25/10%, 20 resamples each; research assistant)	ERA5 + residual OK vs best pure interpolation: 41 km −5.9% (90/136), 53 km −8.8% (107/136), 82 km −13.4% (126/136, where raw ERA5 already beats interpolation)

Date	What Was Done	One-Line Result
2026-10-14	Reviewers E/F audit of S2G, HRRR, and RTMA; Lead check of S2G preprint	S2G's 25–32% gain was evaluated grid-wide against RTMA (non-independent truth), whereas held-out station validation reported only an 8.5% centered-RMSE reduction; HRRR f00 assimilates surface winds; RTMA lacks full-network leave-one-out validation
2026-10-14	r1 Supplement (WorldCover 1 km land-cover stratification)	Urban built-up stations (16): ERA5 + residual 1.530 vs OK 1.523 (no improvement, 9/16); non-built-up (121): 1.556 vs 1.625 (–4.2%, 72/121)

Parameter Provenance Table

Parameter	Value	Source or "Assumed"	Where Checked
Primary Pass Criterion	Fusion / ML reduces RMSE by ≥20% over best pure interpolation AND improves ≥2/3 of stations	Pre-specified in study design (Lead Supplement 2)	—
HRRR Background Lead Time	f01, excluded from primary criterion	f00 assimilates surface station winds (Benjamin 2016, Dowell 2022, verified by reviewer F); f01 still retains memory of assimilated stations	Review
Does ERA5 Assimilate 10 m Land Station Winds?	Not assimilated directly over land	Hersbach et al. 2020 (*QJRMS* 146, 10.1002/qj.3803) observation blacklist, confirmed by reviewers D, A, and E	Review
Node B Holland Chain (Reused in H2)	CLIMADA 6.1.0 H08 + directional marine/land roughness; 2-min / 5-s gust conversion, 500 m gradient reference height	**Assumed**: pre-registered in B prior to execution; all citations of Holland-chain errors carry this caveat	Node B REUSE_NOTES.md
Station Cleaning Rule	Exclude offshore platforms and co-located duplicate coordinates; retain one station per pair within <2 km	Study design (mandatory revision from review)	—

Open Items and Next Steps

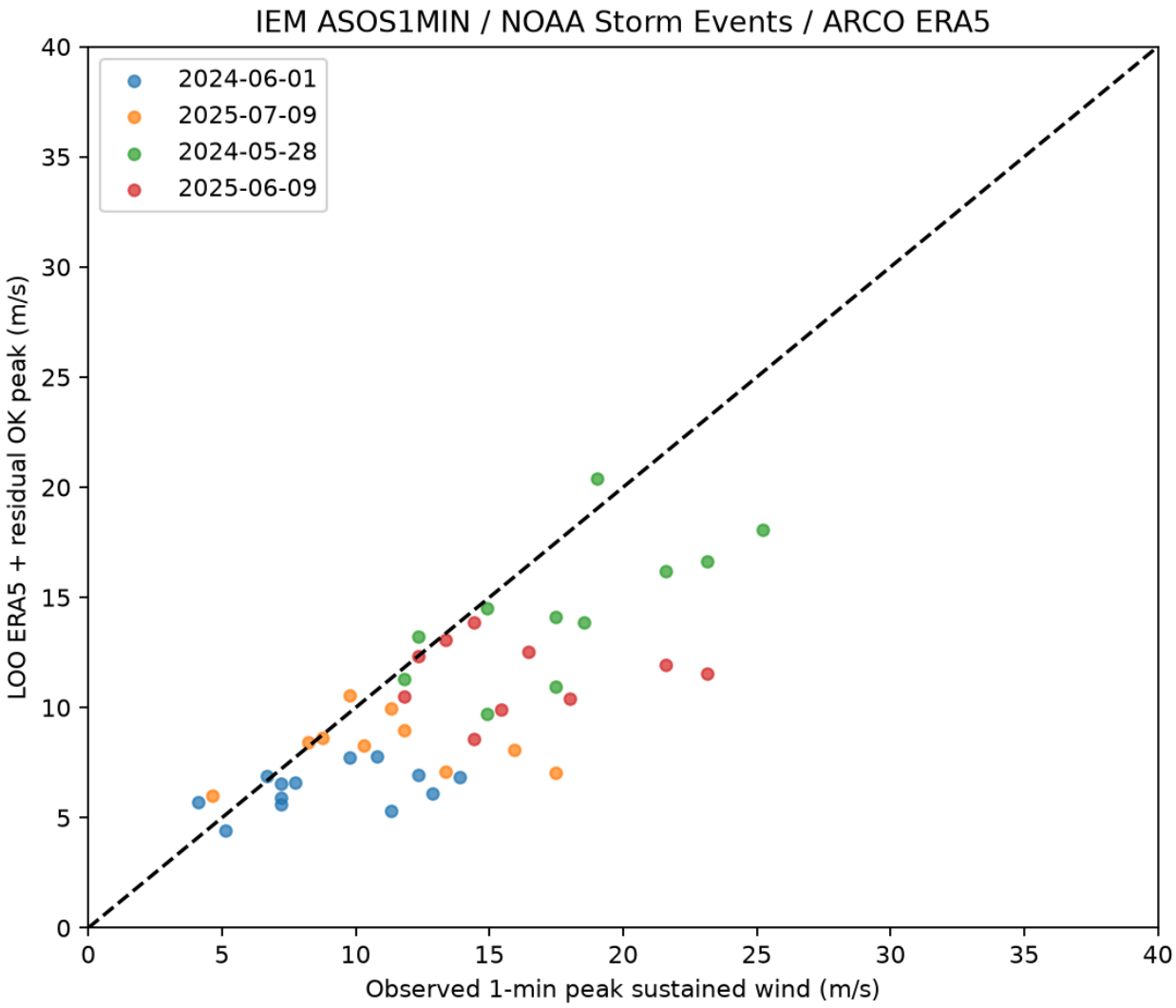
- **H5 peak underprediction correction completed**: Hazard-stratified additive correction proved effective and has been incorporated into Section 4 of Protocol `v1.2` and delivered to Node I via the parent node.
- **Protocol review completed**: Following the review verdict (3/3 conditional support), `v1` was superseded by `v1.1` and then `v1.2` (with `H5` added) and handed over to Node I.
- **H6 & H7 thunderstorm radar pilots completed and wrapped up**: Because full "radar + station" thunderstorm footprint reconstruction (dealiasing, beam-height correction, and case-by-case validation) would require weeks of effort beyond Quick-Trial scope and `H6`/`H7` showed no viable signal on public data, the thunderstorm radar track is closed without a promotion proposal.
- **Node B reuse verification (10-14)**: Confirmed that `H2` imported only the model functions from B without reading uncleaned `v1` root outputs, added Hurricane Francine to align cohorts, and noted survivorship bias at near-eyewall stations due to power/comms dropouts.
- Cross-line coordination: Confirmed zero duplication with both the Event Census line (reusing their read-only `S1_ROWS.csv`, station completeness tables, and sensor notes) and the WRF line (incorporating their notes on HRRR `f01`–`f03` analysis memory, METAR averaging conventions, and land-cover stratification).

Literature Basis

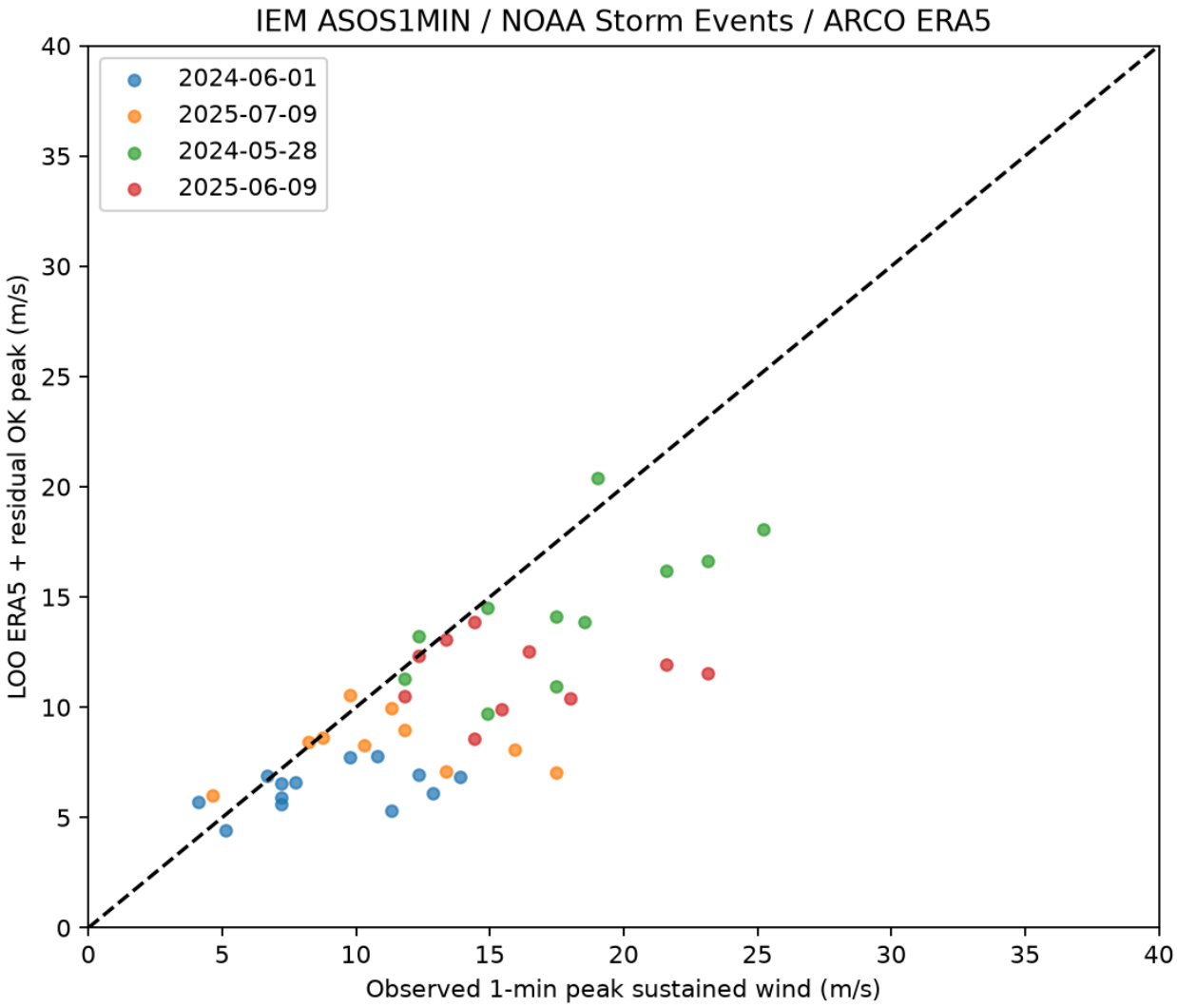
- Zheng et al. 2026, arXiv:2609.11230v1 (`KiloDA`), PDF Fig. 5 text: `−15.4%` RMSE vs Kriging on held-out station block (verified in original PDF by Lead; card pending).
- Kiessling et al. 2021, *Proc. R. Soc. A*, 10.1098/rspa.2021.0236, Table 1: `0.370` vs `0.388` (verified in PMC full text by Lead; card pending).
- `GenDA` evidence boundary card.

Notes from Liang Hu

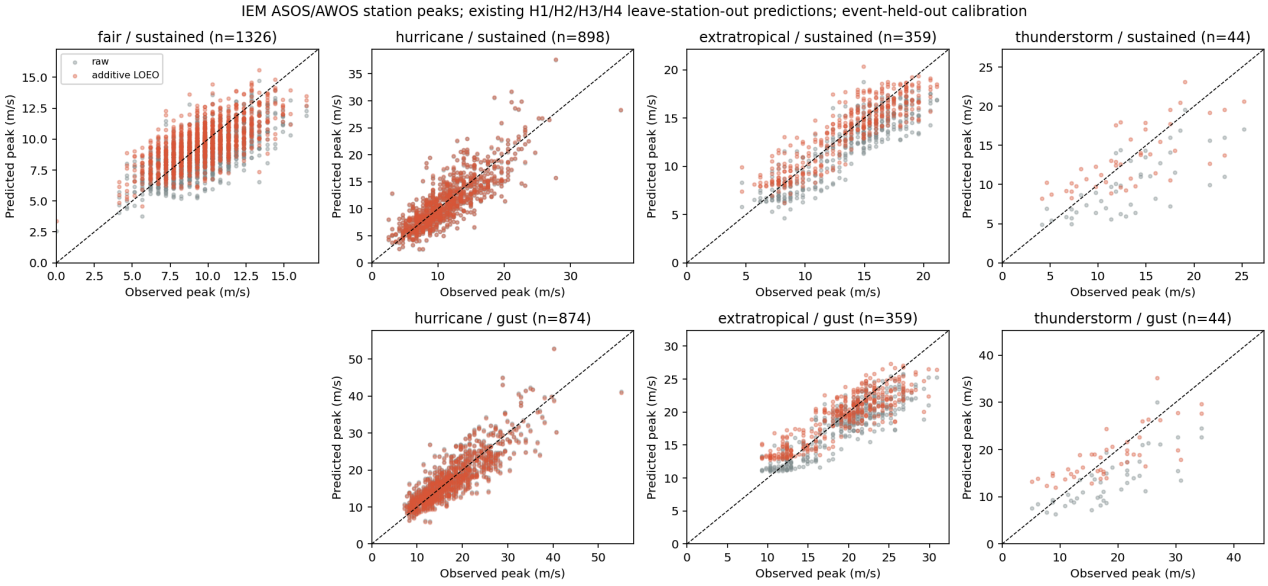
Key figures



peak scatter h4v2

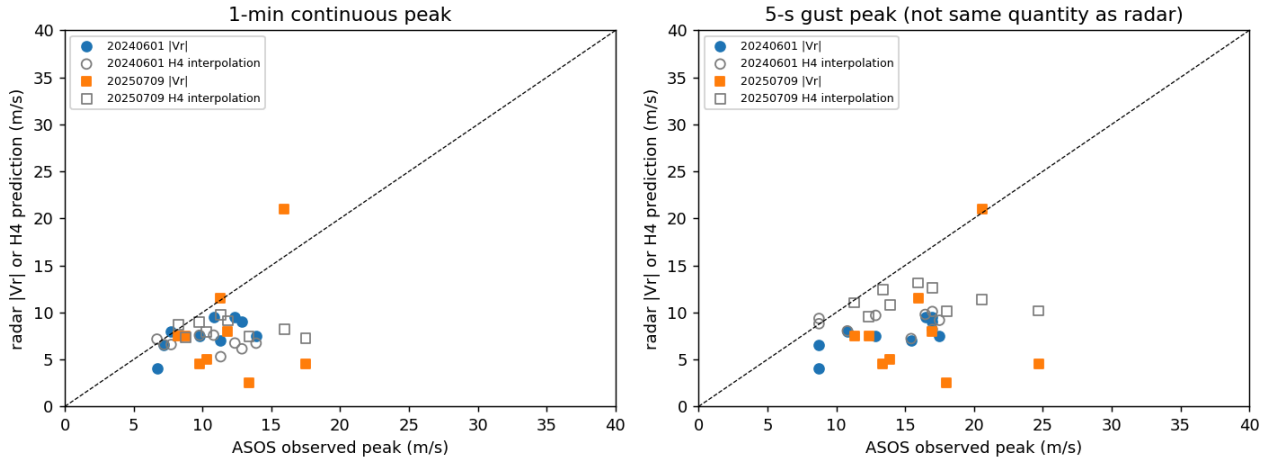


peak scatter h4v3



peak scatter h5

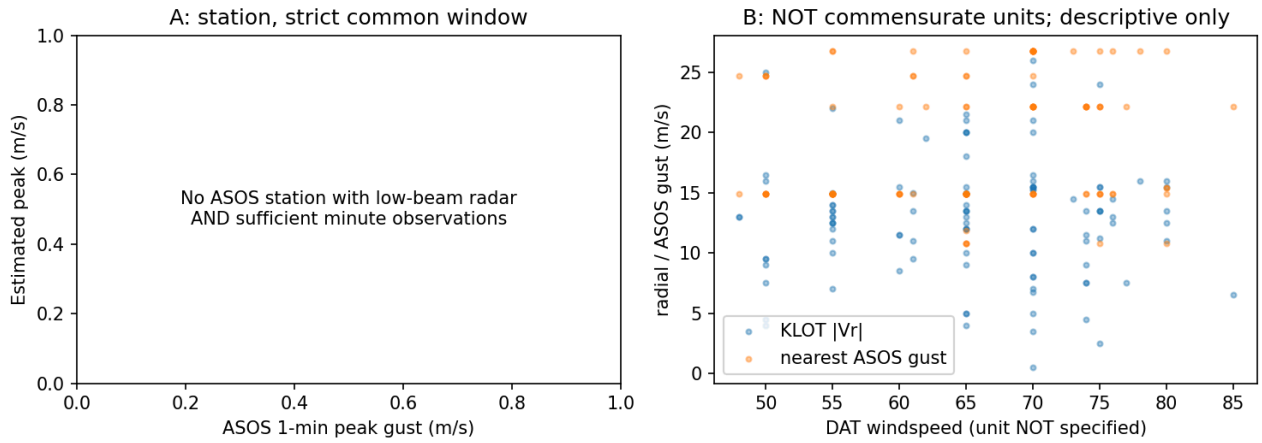
KFWS lowest velocity sweep; radar beam-height radial component is NOT 10-m gust



NOAA NEXRAD Level II public Unidata S3 mirror; IEM ASOS 1-minute; H4 LOO interpolation

radar vs station h6

Sources: NWS DAT FeatureServer, NOAA/Unidata KLOT Level II, IEM ASOS 1-min



radar comparison h7